# Practice workbook · Describe how an impulse crosses a synapse

Naming the synapse, defining a neurotransmitter, and sequencing transmission from an arriving impulse to a new impulse in the next neurone, including through a reflex arc.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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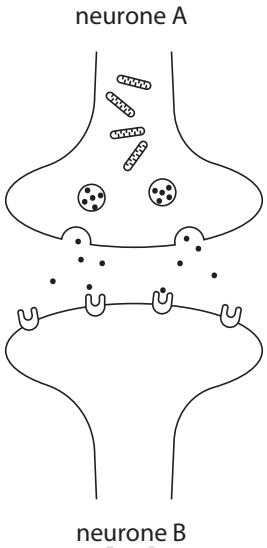
Type ID: HBI-EXPLAIN-SYNAPTIC-TRANSMISSION-SEQUENCE

# 1. Question 3 · part a.iii · 6 marks

November 2024 Paper 2

Worked-example retry · Attempt part a.iii only.

**3** (a) The diagram shows the gap between two neurones, A and B.




(i) Name this gap. **(1)**

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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# 1. Question 3 · part a.iii · 6 marks

November 2024 Paper 2

Worked-example retry · Attempt part a.iii only.

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(iii) Describe how a nerve impulse travels from one side of the gap to the other.

**(6)**

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(iv) Explain why an impulse can only travel in one direction across the gap.

**(3)**

## 1. Question 3 · part a.iii · 6 marks · Working and answer

Working and answer

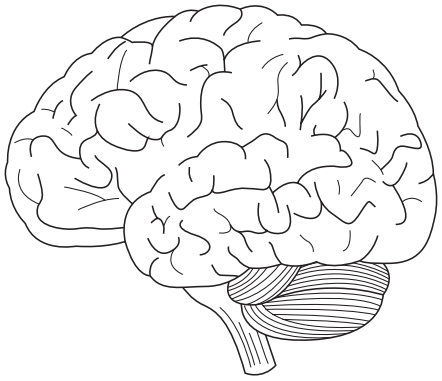
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## 2. Question 7 · part b.iii · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

**7** (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

**(3)**

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

**(1)**

(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

**(1)**

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**20**

## 2. Question 7 · part b.iii · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

(iii) Explain how one neurone communicates with another neurone.

**(3)**

(c) Recent scientific research into the treatment of Parkinson's disease has involved the use of stem cells.

Explain how stem cells could be used to reduce the symptoms of Parkinson's disease.

**(3)**

**(Total for Question 7 = 11 marks)**

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## 2. Question 7 · part b.iii · 3 marks · Working and answer

Working and answer

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# 3. Question 5 · part c.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.ii only.

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**5** This question is about the nervous system.

(a) Describe what is meant by a reflex action.

**(3)**

(b) Diagram 1 shows a reflex arc.

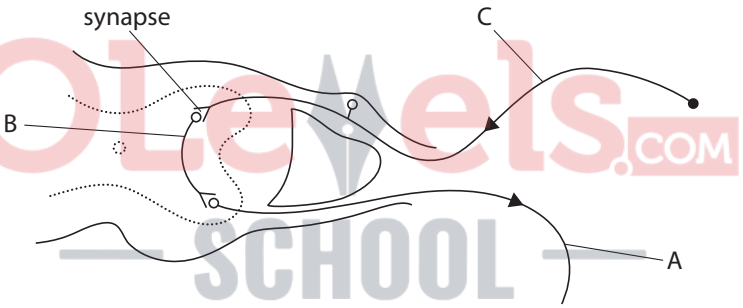



**Diagram 1**

Name the types of neurone labelled A, B and C.

**(3)**

A .....

B .....

C .....

# 3. Question 5 · part c.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.ii only.

(c) Diagram 2 shows a neurone from a healthy brain and a neurone from the brain of someone with Alzheimer's disease.

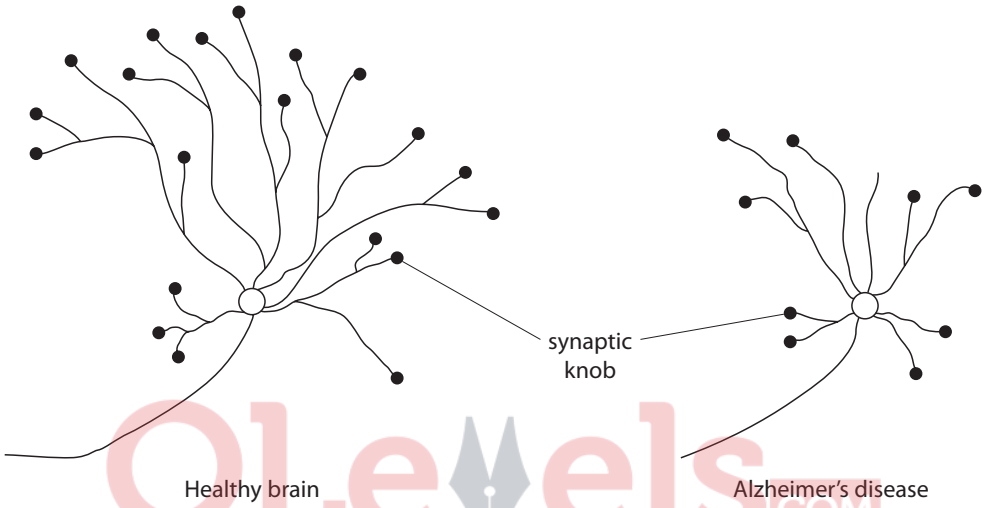



**Diagram 2**

(i) Give three differences between the two neurones. **(3)**

1 .....

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2 .....

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3 .....

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## 3. Question 5 · part c.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.ii only.

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(ii) Acetylcholine is a neurotransmitter.

Explain the importance of acetylcholine in the functioning of the brain.

**(3)**

(iii) One symptom of Alzheimer's disease is that less acetylcholine is produced in the brain.

Explain why there is less acetylcholine produced in people with Alzheimer's disease.

Use information from diagram 2 to help your answer.

**(3)**

**(Total for Question 5 = 15 marks)**

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## 3. Question 5 · part c.ii · 3 marks · Working and answer

Working and answer

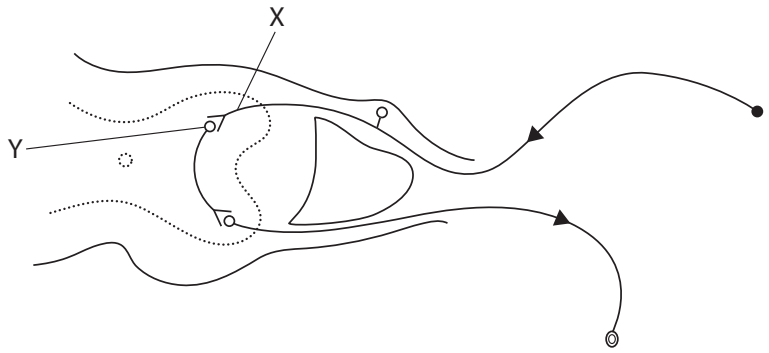
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# 4. Question 8 · part c · 6 marks

June 2021 Paper 1

Further practice · Attempt part c only.

**8** The diagram shows a reflex arc involved in a reflex action.




(a) Describe the function of a reflex action.

**(3)**

(b) Using a line labelled Z, show on the diagram which part of the reflex arc would be connected to a receptor.

**(1)**

(c) Describe how the nerve impulse passes from position X to position Y in the reflex arc.

**(6)**

**(Total for Question 8 = 10 marks)**

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## 4. Question 8 · part c · 6 marks · Working and answer

Working and answer

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## 5. Question 6 · part b.ii · 4 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

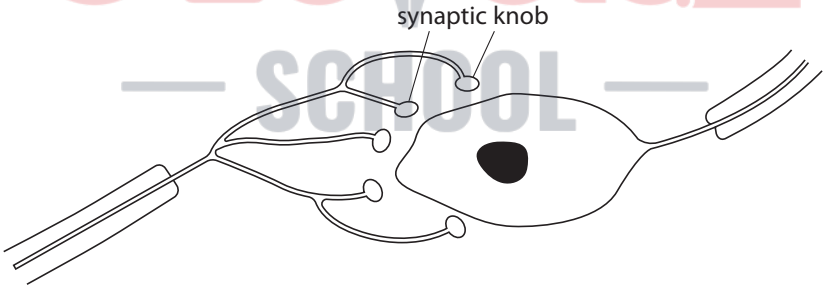
**6** (a) Co-ordination of the activities of the body involves the nervous system and the hormonal system.

Complete the table to compare the two systems.

**(3)**

| | Nervous system | Hormonal system |
|---|---|---|
| **Route of transmission** | | |
| **Type of signal** | | |
| **Relative speed** | | |

(b) The diagram shows a synapse.




(i) Draw an arrow on the diagram to show the direction an impulse travels.

**(1)**

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## 5. Question 6 · part b.ii · 4 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Explain how an impulse travels across a synapse.

**(4)**

(iii) Explain why an impulse can only travel in one direction across a synapse.

**(4)**

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**(Total for Question 6 = 12 marks)**

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## 5. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

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# 6. Question 3 · part b.i · 1 mark

June 2022 Paper 2

Further practice · Attempt part b.i only.

**3** (a) The nervous system is adapted to transfer information rapidly from one part of the body to another.

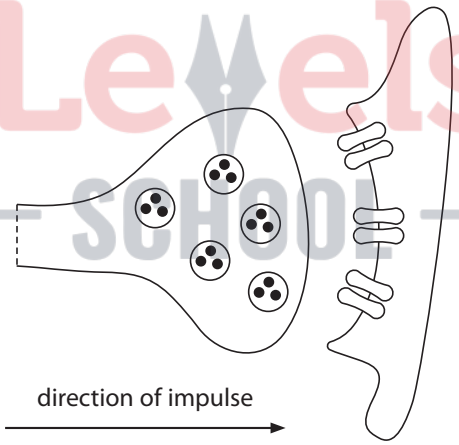
Complete the passage about the nervous system by giving the correct words.

**(4)**

A change in the external environment is detected by ..... in sense organs. These structures convert stimuli into electrical impulses which then travel along ..... neurones to the central nervous system.

The central nervous system coordinates the incoming information using ..... neurones. These neurones then transfer the electrical impulses to ..... neurones.

(b) The diagram shows the junction between two neurones.




(i) State the name of the junction between two neurones.

**(1)**

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## 6. Question 3 · part b.i · 1 mark

June 2022 Paper 2

Further practice · Attempt part b.i only.

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(ii) Describe how an impulse reaching the end of one neurone produces an impulse in the next neurone.

**(2)**

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(c) A nerve impulse travels at a speed of 120 metres per second.

Calculate the distance that this nerve impulse will travel in 5.0 minutes.

**(2)**

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distance = ..... m

**(Total for Question 3 = 9 marks)**

Working and answer

# 7. Question 3 · part b.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

**3** (a) The nervous system is adapted to transfer information rapidly from one part of the body to another.

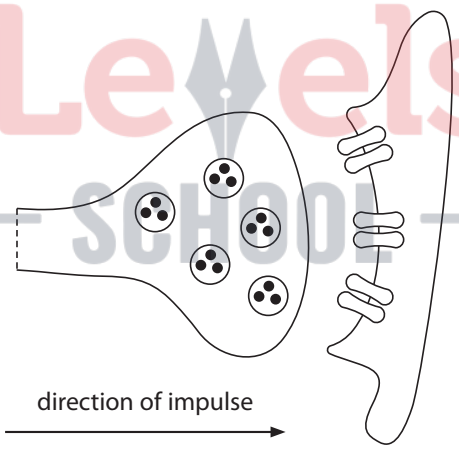
Complete the passage about the nervous system by giving the correct words.

**(4)**

A change in the external environment is detected by ..... in sense organs. These structures convert stimuli into electrical impulses which then travel along ..... neurones to the central nervous system.

The central nervous system coordinates the incoming information using ..... neurones. These neurones then transfer the electrical impulses to ..... neurones.

(b) The diagram shows the junction between two neurones.




(i) State the name of the junction between two neurones.

**(1)**

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## 7. Question 3 · part b.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Describe how an impulse reaching the end of one neurone produces an impulse in the next neurone.

**(2)**

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(c) A nerve impulse travels at a speed of 120 metres per second.

Calculate the distance that this nerve impulse will travel in 5.0 minutes.

**(2)**

distance = ..... m

**(Total for Question 3 = 9 marks)**

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## 7. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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# 8. Question 5 · part a.i · 2 marks

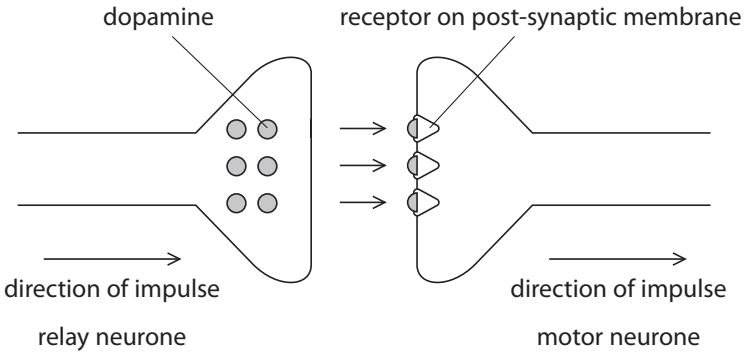
June 2023 Paper 2R

Further practice · Attempt part a.i only.

**5** (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.




(i) Describe what is meant by a neurotransmitter.

**(2)**

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

**(2)**

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## 8. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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# 9. Question 5 · part a.ii · 2 marks

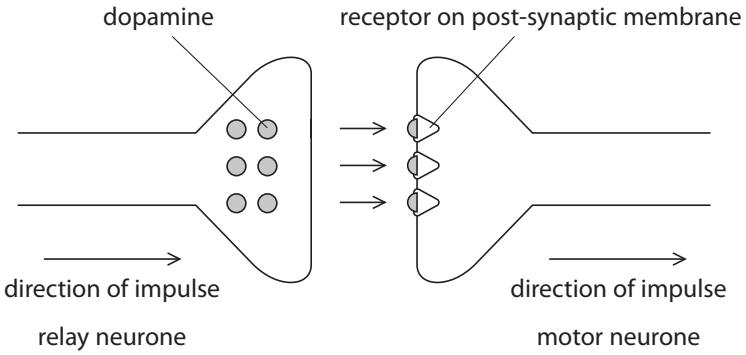
June 2023 Paper 2R

Further practice · Attempt part a.ii only.

**5** (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.




(i) Describe what is meant by a neurotransmitter.

**(2)**

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

**(2)**

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## 9. Question 5 · part a.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

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(iii) Suggest how an excess of dopamine will affect bodily functions.

**(2)**

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.....

(b) In 2017 in the United Kingdom, 145 000 people were diagnosed with Parkinson's disease.

This figure is expected to have an 18% increase by 2025.

(i) Give two symptoms of Parkinson's disease.

**(2)**

1 .....

.....

2 .....

.....

(ii) Calculate the number of people expected to be diagnosed with Parkinson's disease in 2025.

**(2)**

number of people = .....

**(Total for Question 5 = 10 marks)**

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## 9. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

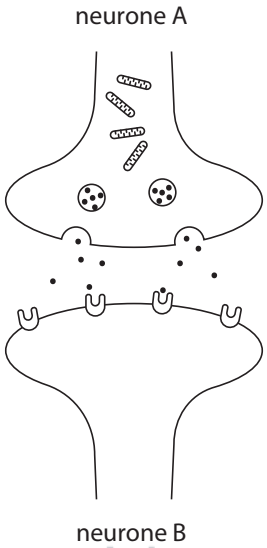
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## 10. Question 3 · part a.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part a.i only.

**3** (a) The diagram shows the gap between two neurones, A and B.




(i) Name this gap. **(1)**

.....

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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## 10. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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